

Sensation

CT

Tune-up System

Tune-up System

This document is valid for:
SOMATOM Sensation 10/16/Cardiac
SOMATOM Sensation 40/64/Cardiac 64/Open

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Safety information

Only qualified and system-trained service staff is allowed to perform CT system installation, service, maintenance and quality assurance. Ensure that the most recent version of the service documentation is available.

To avoid any risk of injury to persons and/or damage to the system, please read and observe the **General Safety Notes** (TD00-000.860.01.xx.xx). Of special importance are the **Product-specific safety notes** (CT02-023.860.01.xx.xx). ([CT02-023.860.01 / Product-specific safety notes](#)) These include highly important safety-related information as well as information about handling screws and nuts, as well as the use of Loctite and instructions for the torque wrench to be applied. You will find the following extract of information in the "Product-specific safety notes":

- Safety information determined on the basis of risk management ([CT02-023.860.01 / Safety warnings](#)).
- General safety information, such as handling of service documentation, CT system training requests, X-ray protection, electrical protection, service tools....([CT02-023.860.01 / General safety information](#)).
- General safety information about working in the gantry with the power off/on ([CT02-023.860.01 / Gantry](#)).
- Use of patient table as lifting device for gantry part replacement ([CT02-023.860.01 / Usage of patient table as lifting device](#)).
- Laser products ([CT02-023.860.01 / Laser Products](#)).
- Attachment of screws and nuts, use of Loctite, adjustment instructions for torque wrench ([CT02-023.860.01 / Screw Connections](#)).

Safety information if automatic movements and/or radiation are in use

Do not leave the system unattended when functions triggering automatic movements and radiation are in use. In this context, please read and observe the following safety information:

CAUTION

[hz_serdoc_F13G01U03M01]

Not following X-rays protective regulations may lead to radiation exposure for yourself and/or other persons.

Risk of radiation exposure!

Observe radiation protection regulations when X-rays are switched on, e.g. :

- ⇒ **Never work inside the gantry room.**
- ⇒ **Do not leave the system unattended.**
- ⇒ **Make sure there is no one inside the gantry room.**
- ⇒ **Make sure that no person can enter the gantry room unnoticed (e.g. lock doors if necessary) .**

⚠ WARNING

[hz_serdoc_F13G01U15M01]

Risk of accident and injury while a program including automatic movements (e.g. automatic table feed or gantry rotation) is running.

Risk of accident and injury!

- ⇒ Do not leave the system unattended.
- ⇒ Make sure there is no one inside the gantry room.
- ⇒ Make sure that no person can enter the gantry room unnoticed (e.g. lock doors if necessary).

Use and handling of phantoms

Only phantoms delivered with the Sensation system can be used for system adjustments. **Other phantoms will definitely generate incorrect adjustment values.**

The water phantom has to be stored at a constant temperature of $+21^{\circ} \pm 3^{\circ}$ Celsius! For measurements, the water temperature in the phantom has to be $+21^{\circ} \pm 3^{\circ}$ Celsius!

The density of the water in the phantom depends on the water temperature. Changing the water temperature by $+1^{\circ}$ Kelvin corresponds to changing the water equivalent by 0.3 HU (Hounsfield unit) in the CT image. **In this context, refer to the following safety information**

⚠ CAUTION

[hz_serdoc_F13G10U01M01]

Poor image quality through wrong adjustment values may be created during tune up.

Risk of misalignment!

- ⇒ Only phantoms delivered with the CT system may be used for system adjustments.
- ⇒ The water phantom has to have the same temperature as the examination room.

Structure of this document

This document is divided into three parts:

- **Installations**

Guided tours: Predefined sequences of tune-up steps after replacing/installing the parts specified.

- **Adjustments**

Detailed description of each single adjustment step.

- **Table generation**

Detailed description of each single table generation step.

Phantoms & Tools

The tune-up on this system requires new phantoms and a new phantom holder to ensure a defined vertical position of the phantom. **Please keep in mind, only phantoms or tools delivered with the system may be used for the tune-up. Other phantoms will definitely generate false adjustment values.** Do not mix the phantoms with other system phantoms, specifically those for the Somatom Volume Zoom and Sensation 4. The following phantoms & tools are required to perform the procedures described in this service document:

Tab. 1 Sensation 10/16/Cardiac16/40/64/Cardiac64/Open

Designation	Part no.	Explanation
Combi phantom	73 96 653	Part of the system phantom set. Used for CCR head, CCR body, spacing, focus alignment
Combi phantom (only for Sensation 40/64/Cardiac64/Open)	88 72 074	Part of the system phantom set. Used for CCR, spacing, focus alignment
20 cm water phantom	48 06 977	Part of the system phantom set. Used for image quality, tune-up
Wire phantom	48 06 951	Part of the system phantom set. Used for image quality, tune-up
Slice thickness phantom	73 96 604	Part of the system phantom set. Used for image quality, Tune-up
Alignment phantom	73 96 901	Part of the system phantom set. Used for focus alignment
Phantom holder	73 96 943	Part of the system phantom set. Used for all Sensation 16 phantoms only
Bracket	30 62 242	Delivered with the system (table). Horizontal adjustment of tabletop
Pin	30 62 283	Delivered with the system (table). Horizontal adjustment of tabletop
Gantry balancing set	73 96 091	Delivered with the system Set of balancing weights for mechanically balancing the gantry.

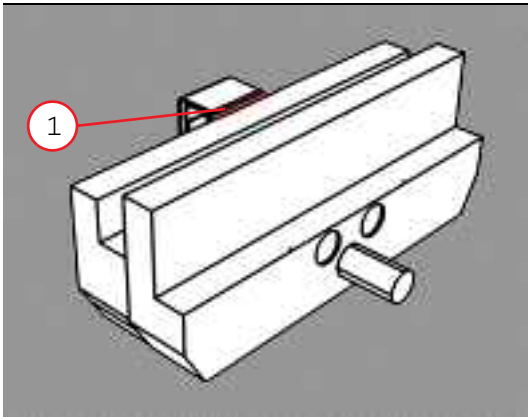


Fig. 1: Combi phantom
Pos. 1 Starting point

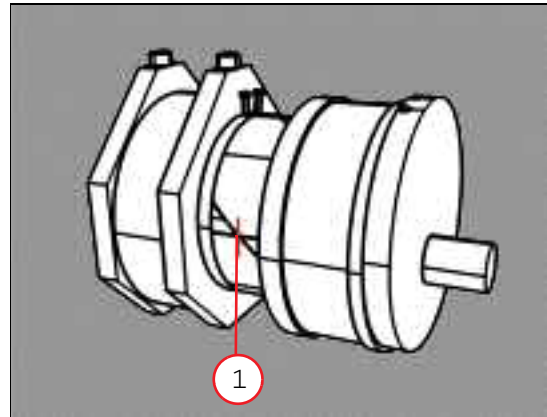


Fig. 2: System phantom
Pos. 1 Crossover point

NOTE

Fig.1 shows the combi phantom used for Sensation10/16/Cardiac. The combi phantom used for Sensation40/64/Cardiac64/Open consists of the body CCR phantom only, the head CCR bar is no longer available.

NOTE

With Somaris VB20 a new system phantom is available. The new system phantom contains a new high resolution phantom (wire in plexi). Somaris VB20 still supports the old system phantom (old high resolution phantom- wire in air).

Acronyms and Abbreviations

The following acronyms and abbreviations are used in this document:

Tab. 2 Acronyms and Abbreviations

ADU	A cquisition C ontrol & D ose M odulation U nit, PCB within the C-box of the system
ADM	A rray D etector M odule (Detector Module), PCB within the detector
CAN	C ontroller A rea N etwork
DMS	D ata M easuring S ystem
DPU	D igital P ower U nit
FOC	F Ocus d eflection and C olimator control, PCB within the C-box of the system
FW	F irm W are
HU	H ounsfield U nits
HV	H igh- V oltage
HW	H ard W are
ID	I Dentification
IRS	I mage R econstruction S ystem
IMS	I mage M easuring S ystem
MTF	M odulation T ransfer F unction
n.a.	not applicable
PCB	P rinted C ircuit B oard
PDC	P ower D istribution C abinet
SW	S oft W are
tbd or TBD	to be defined
UHR comb	U ltra H igh R esolution comb
WCS	W ater C ooling S ystem

General

Safety

WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.**

Phantoms

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used.

Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open:



Fig. 3: Phantom holder

Overview

Guided Tours

The software provides a Guided Tour (Sequence) for the following FRU's:

- Sequence after tube replacement

- Sequence after replacing tube collimator box
- Sequence after HV tank, D400 or E-Box replacement
- Sequence after detector replacement
- Sequence after module (peripheral) replacement
- Sequence after module (central) replacement
- Sequence after reference monitor replacement (only for Sensation10/16/Cardiac16)
- Sequence after reference monitor board replacement (only for Sensation40/64/Cardiac64/Open)
- Sequence after complete UHR assembly replacement (only for Sensation10/16/Cardiac16/40/64/Cardiac64)
- Sequence after FOC board D514 replacement
- Sequence after PHI_Z_MON replacement

Expert Mode

The experienced CSE can perform single tuneup or adjustment steps to correct a system malfunction. Therefore each tuneup and adjustment program can be started within the Expert Mode.

Save system table

At the end of the Guided Tour a backup of the files is requested.

Start <Backup & Restore> from the home menu of the service software

Select "Backup" in the left frame

Select a drive in the drop-down menu according to the local backup method, usually involving MOD or CD-R

Select the package "CT scanner" within the drop-down menu

Click <GO> to start the procedure

Check the screen output for the success message

Report function

Under **Local_Service --> Reports (upper navigation bar)--> Tune-Up/QA/Tests --> Tuneup** the results of the executed tuneup steps are displayed.

All tune-up steps shall be completed successfully without errors and inside the given tolerances.

If any of the tune-up routines does not finish successfully or if the result is unsatisfactory, the procedure has to be repeated.

Sequence after tube replacement

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **X-ray Tube TuneUp**.

The following tune-up steps are called up automatically, one after the other:

- Configuration
- Balance gantry
- Getter
- ZAdjust
- FocusAli
- ZFocusCheck (only for Sensation40/64/Cardiac64/Open)
- Insert plexi strip
- GenAdapt
- Air calibration
- Water scaling
- Final work (quality test, save tables)

Sequence after replacing tube collimator box

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Tube Collimator**.

The following tune-up steps are called up automatically, one after the other:

- Configuration
- ZAdjust
- FocusAli
- Insert plexi strip
- GenAdapt
- Air calibration
- Water scaling
- Final work (quality test, save tables)

Sequence after HV tank, D400 or E-Box replacement

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **HV tank, D400**.

The following tune-up steps are called up automatically, one after the other:

- Configuration
- GenOsc
- Insert plexi strip

Sequence after detector replacement

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Detector**.

The following tune-up steps are called up automatically, one after the other:

- Configuration
- Balance gantry
- Remove def. channels
- ZAdjust
- Defective channels (from software version VB10 onward)
- FocusAli
- ZCombAli (only for Sensation64/Cardiac64 with UHRplus only)
- CombAli
- ZFocusCheck (Sensation40/64/Cardiac64/Open)
- Insert plexi strip
- GenAdapt
- Defective channels (in software version VA70)
- Channel correction
- Move plexi strip gap
- Air calibration
- Spacing
- Water scaling
- Final work (quality test, save tables)

Sequence after module (peripheral) replacement

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Module (periph.)**.

The following tune-up steps are called up automatically, one after the other:

- Module Z-Align. - Functional test (only for Sensation 40/64/Cardiac64/Open)
- Check Module Align. - Guided Module Alignment only if Module Alignment test is out of tolerance.
- Insert plexi strip
- Defective channels
- Channel correction
- Move plexi strip gap
- Air calibration
- Spacing
- Final work (quality test, save tables)

Sequence after module (central) replacement

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Module (central)**.

The following tune-up steps are called up automatically, one after the other:

- Module Z-Align. - Functional test (only for Sensation 40/64/Cardiac64/Open)
- Check Module Align. - Guided Module Alignment only if Module Alignment test is out of tolerance.
- Insert plexi strip
- Defective channels
- GenAdapt
- Channel correction
- Move plexi strip gap
- Air calibration
- Spacing
- Water scaling
- Final work (quality test, save tables)

Sequence after reference monitor replacement (only for Sensation 10/16/Cardiac16)

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Reference Monitor**.

The following tune-up steps are called up automatically, one after the other:

- Insert plexi strip
- GenAdapt
- Air calibration
- Water scaling
- Final work (quality test, save tables)

Sequence after reference monitor board replacement (only for Sensation 40/64/Cardiac64/Open).

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Reference Monitor Board**.

The following tune-up steps are called up automatically, one after the other:

- Insert plexi strip
- Air calibration
- Final work (quality test, save tables)

Sequence after complete UHR assembly replacement (Sensation10/16/Cardiac16/40/64/Cardiac64)

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Complete UHR Assembly**.

The following tune-up steps are called up automatically, one after the other:

- Balance gantry
- ZCombAli (only for Sensation64/Cardiac64 with UHRplus)
- CombAli
- Insert plexi strip
- Air calibration
- Water scaling
- Final work (quality test, save tables)

Sequence after FOC board replacement

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **Module (central)**.

The following tune-up steps are called up automatically, one after the other:

- FocusAli
- Insert plexi strip
- Final work (quality test, save tables)

Sequence after PHI_Z_MON replacement

For this sequence, an automated guided tour is available.

- Select **Tune-up**.
- Select **PHI_Z_MON**.

The following tune-up steps are called up automatically, one after the other:

- ZAdjust
- FocusAli
- Insert plexi strip
- GenAdapt
- Air calibration
- Water scaling
- Final work (quality test, save tables)

General

Safety

WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.

Phantoms

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used.

Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open:



Fig. 4: Phantom holder

Overview

The following adjustments are available:

- Balancing the gantry
- Horizontal table movement
- Vertical table movement

- Oscillator frequency of the generator
- X-ray tube with the “getter” procedure
- Z-position of the tube diaphragm
- Focus alignment
- High-resolution UHRplus comb
- High-resolution comb
- Generator adaptation
- Filament adaptation
- Light marker

Additionally, the <Tune-up> procedure allows you to check the following adjustments:

- Checking the Z-position of the tube diaphragm
- Checking the focus alignment
- Checking the focus spot defelection in phi- and z-direction.

Save system table

Create a backup of the files after the tune-up procedure.

Start <Backup & Restore> from the home menu of the service software

Select “Backup” in the left frame

Select a drive in the drop-down menu according to the local backup method, usually involving MOD or CD-R

Select the package “CT scanner” within the drop-down menu

Click <GO> to start the procedure

Check the screen output for the success message

Report function

Under **Local_Service --> Reports (upper navigation bar)--> Tune-Up/QA/Tests --> Tuneup** the results of the executed tuneup steps are displayed.

All tune-up steps shall be completed successfully without errors and inside the given tolerances.

If any of the tune-up routines does not finish successfully or if the result is unsatisfactory, the procedure has to be repeated.

Balancing the gantry

General

Improved temporal resolution in CT cardio images was achieved by increasing the gantry rotating speed. At certain points of the gantry, the centripetal forces are more than 20g. This means a 25g weight difference will result in a 500g weight difference at the highest rotation speed. For this reason, a balancing device is integrated into the gantry, even though each system is accurately balanced prior to delivery.

An automated 2-plane balancing procedure is implemented within the <Tune up> of the service software.

Safety

WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.**

Check measurement of the “Balance Gantry”

NOTICE

The gantry doors must be closed and fixed by the door screws during the measurement steps.

- ⇒ **Avoid mismeasurement of balancing, due to wobbling gantry doors.**

- **Only for Somaris < VB10.** Starting with VB10 the fans are switched off by software. **Switch off the gantry fans** (F304, F305 and F308 in left gantry stand).
- Close the front and cover of the gantry and insert the plexi strip.
- Change to the service mode **Options / Local Service**.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Balance Gantry**.
- **Do not select** “include calibration”
- Select **Go**.
- Press **START**.

Wait until the message "**Balance gantry completed**" appears.

NOTE

Only if the check measurement is not successful:

The next step “Adjustment procedure of the Balance Gantry” should be carried out.

- **Switch on the gantry fans only for Somaris < VB10.** (F304, F305 and F308 in left gantry stand)

Adjustment procedure of the “Balance Gantry”

Prerequisites

Tab. 3 Sensation 16: Phantoms and tools required for the tune-up

Designation	Part no.	Explanation
Gantry balancing set	7396091	Delivered with the system Set of balancing weights for mechanically balancing the gantry.

Adjustment procedure

- **Only for Somaris < VB10.** Starting with VB10 the fans are switched off by software. **Switch off the gantry fans** (F304, F305 and F308 in left gantry stand).
- Change to the service mode **Options / Local Service**.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Balance Gantry**.
- Select **“include calibration”**
- Select **Go**.
- Press **START**.

Calibration of the gantry:

Follow the program output instructions and insert/remove the calibration weights accordingly!

NOTE

Tighten the screws of the front balance weights with a torque of 10 Nm. The screws must be secured with Loctite 243.

Tighten the screws of the backside balance weights with a torque of 25 Nm. The screws must be secured with Loctite 243.

The last element of the balance weight package must consist of steel (disk or pressure plate).

There is still power at the slip ring even if the gantry is switched off (detector heating).

Before installing the weights near the slip ring, switch off fuses F8, F9 and F12.

After installing the weights, do not forget to switch on the fuses again.

After switching on the gantry it takes a while until the gantry is in stand by mode. Continue with the balancing procedure after the initialization has been completed.

Basically, 3 measurements will be performed:

- Reference measurement - no calibration weights installed
- Calibration of plane 1 (back side) - with calibration weights installed
- Calibration of plane 2 (front side) - with calibration weights installed

Wait until the message "**Calibration completed**" appears.

Balancing the gantry:

Follow the program output instructions and insert/remove the weights accordingly!

NOTE

Tighten the screws of the front balance weights with a torque of 10 Nm. The screws must be secured with Loctite 243.

Tighten the screws of the backside balance weights with a torque of 25 Nm. The screws must be secured with Loctite 243.

The last element of the balance weight package must consist of steel (disk or pressure plate).

There is still power at the slip ring even if the gantry is switched off (detector heating).

Before installing the weights near the slip ring, switch off fuses F8, F9 and F12.

After installing the weights, do not forget to switch on the fuses again.

After switching on the gantry it takes a while until the gantry is in stand by mode. Continue with the balancing procedure after the initialization has been completed.

Explanation:

The unbalancing measurement of the figure shows the absolute position and weight of unbalance. Please note, the gantry only has a limited amount of positions for the compensation weights. For this reason, the program has to calculate the size and position of the compensation weights for the gantry. These positions are shown in the compensation instructions according to the label on the gantry. When the compensation has been performed and accepted by the user, the information will be stored in a table. The subsequent <Balance Gantry> is based on the assumption that the weights have been entered and inserted correctly!

NOTE

All previous actions regarding <Balance Gantry> are stored by the system. For this reason, the screen layout may differ! The option "Remove weights from gantry" appears only, if sufficient balancing weights are mounted in your system!

Wait until the message "**Balance gantry completed**" appears.

- **Switch on the gantry fans only for Somaris < VB10.** (F304, F305 and F308 in left gantry stand)

Horizontal table movement

General

The horizontal table movement is mechanically connected to a potentiometer. Any change with regard to horizontal table movement also changes the resistance of this potentiometer. For this reason, the converted ADC values of certain table positions have to be calibrated and stored as a SW table after successfully performing the <PHSHoriz> procedure.

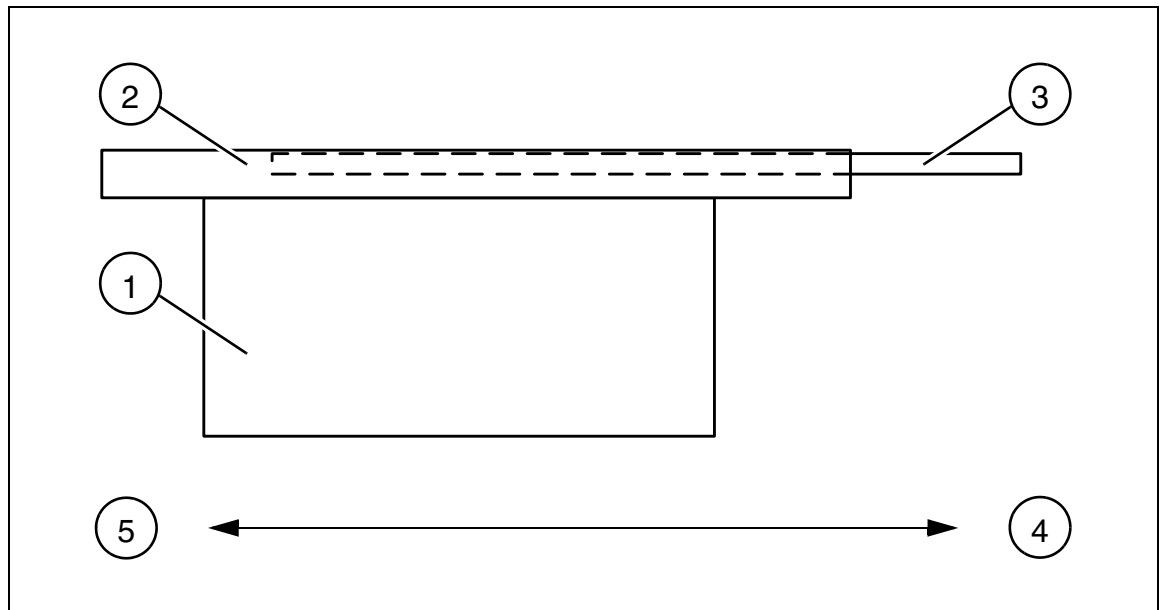


Fig. 5: Overview of the table components to be adjusted

- | | |
|--------|----------------------------------|
| Pos. 1 | Table base |
| Pos. 2 | Top support |
| Pos. 3 | Tabletop |
| Pos. 4 | Horizontal tabletop movement IN |
| Pos. 5 | Horizontal tabletop movement OUT |

The <PHSHoriz> adjustment procedure refers to the adjustment of two components, namely:

- the tabletop, and
- the top support

The tabletop or TOP SUPPORT movements are adjusted via two separate functions shown in the <PHSHoriz> platform. However, the individual adjustment steps are identical, and are described accordingly as follows:

Safety**WARNING**

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.**Risk of accident and injury!**

- ⇒ **Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.**

Prerequisites**Tools and auxiliary equipment**

Tab. 4 Sensation 16/40/64/Open: Phantoms and tools required for the tune-up

Designation	Part no.	Explanation
Bracket	30 62 242	Delivered with the system (table). Horizontal adjustment of tabletop
Pin	30 62 283	Delivered with the system (table). Horizontal adjustment of tabletop

Procedure

- Change to the service mode **Options / Local Service**.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **PHSHoriz**.
- Select **top support** and **tabletop**.
- Select **Go**.
 - ⇒ Follow the program output instructions or the “detailed tabletop adjustment description” below.

NOTE**The tabletop or the top support will move automatically.**

Wait until the message "**PHS Horizontal adjustment completed**" appears.

- Select **Go** (to save the result)

NOTE

It takes approximately 1.5 minutes to store the table and complete a new “init”. During this process, all gantry panel displays are dimmed.

- ⇒ The procedure is completed, if the “init” of the control unit was performed successfully. (All lights on the gantry panel displays should now be on again.)

Detailed tabletop adjustment description

This detailed description provides some helpful hints, especially when performing the <PHSHoriz> tabletop calibration for the first time. For experienced users, the program output should be sufficient.

Description:

The adjustment for the tabletop is performed in two steps. In the first step, the number of encoder pulses is determined for a fixed distance. This distance (1200 mm) is indicated by two holes inside the tabletop frame. For this reason, the tabletop (foot end) has to be moved to this two holes. A bracket and a pin are used to guarantee the accuracy of the calibration.

In the second step, the potentiometer values are calibrated. The tabletop is moved 1650 mm in increments of 50 mm, while the potentiometer value for each increment is stored.

Detailed tabletop calibration description

1. Start a service session, activate the <Tune-up> platform and run the <PHSHoriz> adjustment procedure.
2. Raise the patient table to the vertical table position: 125
3. Select “Table top” in the parameter frame and click <Go> to start the procedure
4. The cover of the table frame has to be removed. For this reason,
 - move the tabletop fully into the gantry funnel
 - remove the 6 hexagon socket screws of the table frame cover
 - remove the back table cover by loosening the 2 hexagonal socket screws
 - shift the table frame cover approx. 5 cm to the back side
 - disconnect protective ground connector PE 715
 - the cover of the table frame is now ready to be removed towards the backside
 - Move the tabletop completely out of the gantry

5. The following pictures are intended to illustrate the screen output for the adjustment procedure.

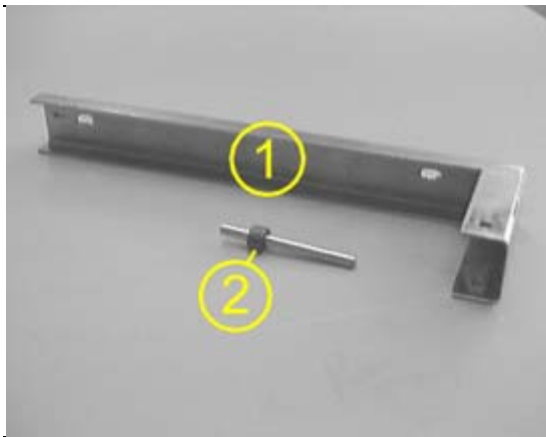


Fig. 6: Bracket and pin

Install the bracket with 2 screws at the back end of the tabletop (Fig. 7 / p. 34).

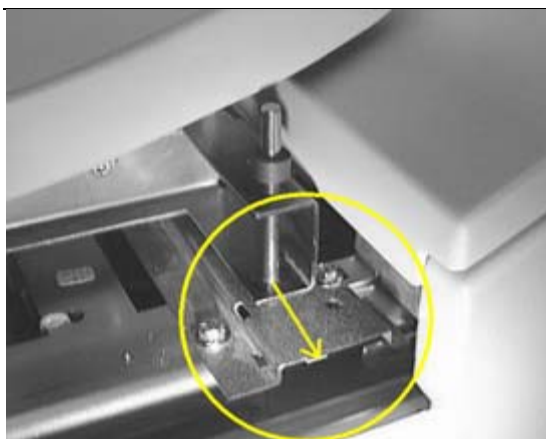


Fig. 7: Table top adjustment, 1st position (rear end)

- Insert the pin with the long side downward in the holes of the bracket
- Move the tabletop slowly toward the gantry until the flat side of the pin touches the back mark of the table frame:
 - ⇒ Ensure that the pin has contact with the back mark.

6. Click **<Go>** as displayed by the program. The table is now released.
7. To calibrate the second position

- Lift the pin and move the tabletop manually towards the gantry
- After 1200 mm a second hole can be found in the table frame
- Insert the pin again in the holes of the bracket (long side downwards, flattened side of the pin facing the gantry)



Fig. 8: Table top adjustment, 2nd position (+1200 mm)

- Move the tabletop slowly toward the gantry until the pin touches the mark of the table frame:

⇒ Ensure that the pin has contact with the back mark.

8. Click <Go> as displayed by the program and remove the pin
9. Move the tabletop out of the gantry until the mechanical end stop is reached.
Move the table 7.5 mm towards the gantry as displayed.
⇒ Use the gantry panel to measure the distance of 7.5 mm.
10. Click <Go> as displayed by the program. The tabletop moves automatically. Do not leave the system unattended when functions triggering automatic movements and radiation are in use. **In this context, please read and observe the following safety information:** ([Safety information if automatic movements and/or radiation are in use / p. 6](#))
11. Wait until the program outputs the message "PHS Horizontal adjustment completed". The new values can be stored by clicking <GO>.

NOTE

It takes approximately 1.5 minutes to store the table and complete a new "init". During this process all gantry panel displays are dimmed.

- ⇒ The procedure is completed, if the "init" of the control unit was successfully performed. (All lights on the gantry panel displays should now be on again.)
- 12. Move the tabletop fully into the gantry funnel. Install the cover of the table frame again. Tighten the 6 hexagonal socket screws of the cover.
⇒ The procedure is now completed.
- 13. Install the covers. For this purpose,

- move the tabletop fully into the gantry funnel
- insert the table frame cover from the backside and connect protective ground connector PE 715
- insert the 6 hexagonal socket screws of the table frame cover
- install the end cover for the back table and insert the 2 hexagonal socket screws
- Move the tabletop completely out of the gantry

Vertical table movement

General

The vertically table movement is mechanically connected to a potentiometer. Any change with regard to vertical table movement also changes the resistance of this potentiometer. For this reason, the converted ADC values of certain table positions have to be calibrated and stored as a SW table after successfully performing the <PHSVert> procedure.

Please refer to the definition of the patient table components and the location of service switches required for the adjustment procedure.

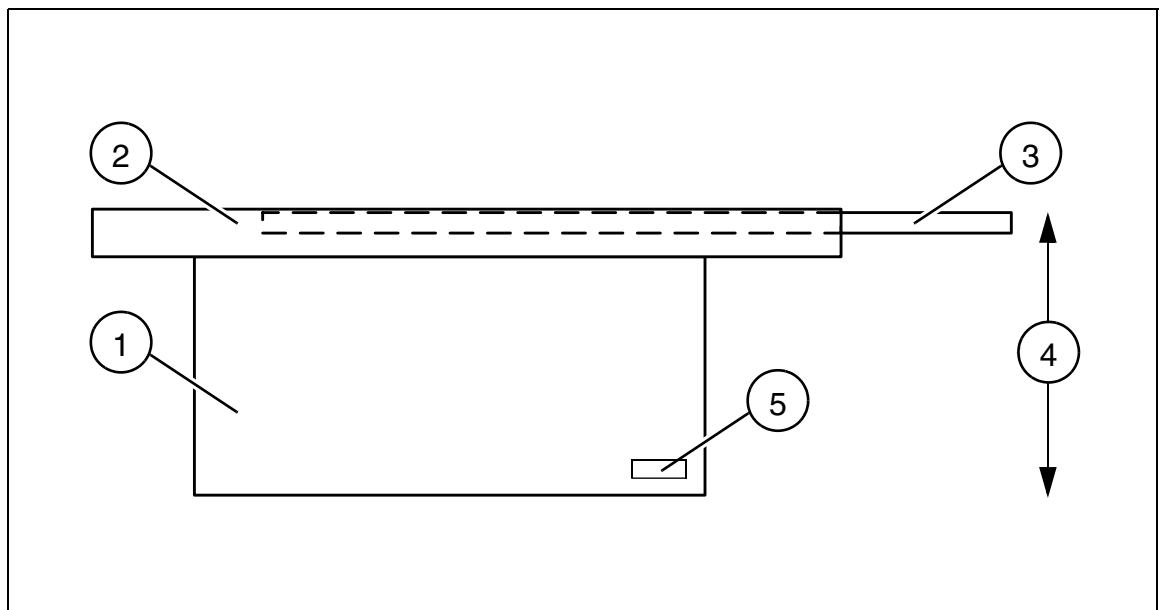


Fig. 9: Overview of the table components and the location of service switches

- Pos. 1 Table base
- Pos. 2 Top support
- Pos. 3 Table top
- Pos. 4 Vertical table movement
- Pos. 5 Service switches (S701, S702, S703)

Safety



[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.

Prerequisites

Tools and auxiliary equipment

Designation	Part number
Tape measure	n.a.

Procedure

NOTE

Prior to adjusting the vertical table travel, move the tabletop and top support into the park position to avoid collisions during the procedure.

Ensure that there is no weight on the tabletop.

- Change to the service mode **Options / Local Service**.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **PHSVert**.
- Select **Go**.

⇒ Follow the program output instructions

NOTE

Wait about 5 seconds before pressing the up or down button after switching to the service function.

Wait until the message "**PHS Vertical adjustment completed**" appears.

- Select **Go** (to save the result)

NOTE

It takes approximately 1.5 minutes to store the table and complete a new "init". During this process all gantry panel displays are dimmed.

⇒ The procedure is completed, if the "init" of the control unit was performed successfully. (All lights on the gantry panel displays should now be on again.)

Adaptation of the generator oscillator frequency

General

The components of the

- HV tank (H1)
- D400
- Generator electronic box

comprise a resonance circuit. As a result, it is necessary to determine the system-specific resonance frequency of these components. For this reason, the generator electronics (GEL) adjust to the optimal oscillator frequency during this procedure. The result is stored in the XRS controller.

When replacing one of the above components, you have to repeat the adjustment procedure.

Procedure

- Remove all absorbers from the measurement field.
- Change to the service mode **Options / Local Service**.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **GenOsc**.
- Select **Go**.
- Press **START**.

The adjustment uses a special static mode and takes approximately 80 sec.

- The message "**Generator Oscillatory Frequency completed**" appears.

Tube gettering

General

Gettering is required after the X-ray tube is replaced or in response to frequent tube arcing.

The getter procedure is used to:

- Check the HV behavior of the X-ray tube
- Improve the HV stability of the X-ray tube, if insufficient
- Guarantee the HV stability of a new X-ray tube

This procedure is also known as “conditioning the X-ray tube”. After you start the procedure with the voltage selected, the program automatically performs a specific number of scans. If the tube arcs, the auto repeat function attempts to repeat the scans to obtain a specific number of scans without arcing. If the tube does not arc during these scans, the voltage is automatically switched to the next higher kV step. Gettering is completed when the highest kV step has been performed without tube arcing.

Procedure

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Getter**.
- Select **Start voltage (kV): 80 kV**.
- Switch on "**Auto repeat after tube arcing**".
- Select **Go**.
- Press **START**.
- When all the steps of the gettering procedure have been completed without errors, the message "**Gettering completed**" appears.

Z position of the tube collimator

General

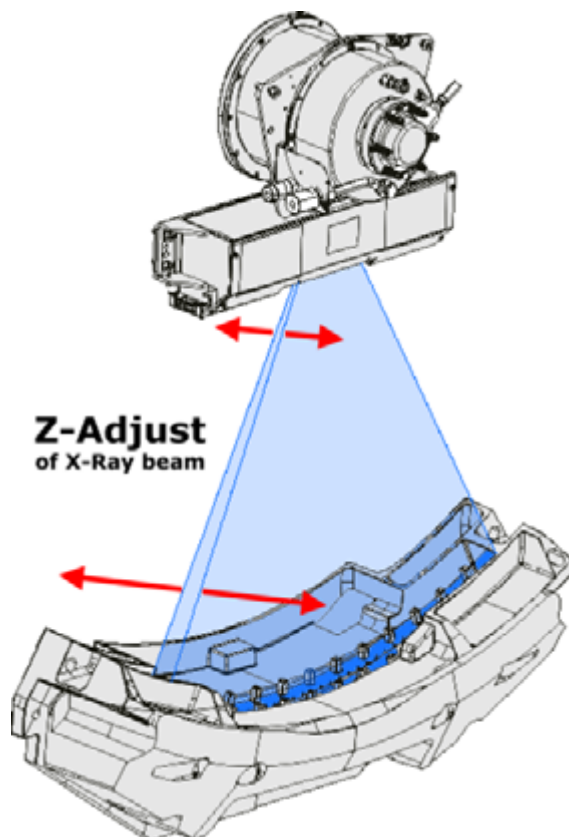


Fig. 10: Adjust the Z-position of the tube diaphragm

To ensure a maximum output signal for the detector elements, the X-ray beam has to be centered precisely over the detector.

The adjustment starts with a defined movement of the tube-side diaphragm toward the back of the gantry. Measurements are taken with each position increment in the opposite direction. The optimal tube diaphragm positions for the low, middle, and high channels of the detector are determined in static mode (no rotation). In addition, the optimal position for two rotation times are determined. During the adjustment, the diaphragm blades are moved accordingly. Since the two blades have separate motors, the diaphragm aperture and the Z-position of the diaphragm can be set simultaneously.

Prerequisites

Tools and auxiliary equipment

2 dial gauges (only if a mechanical adjustment of the tube diaphragm is required by software).

Procedure

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **ZAdjust**.
- Select **Go**.

- Press **START**.
Follow the program output instructions.
If a mechanical adjustment is required: Refer to the information below.
- After the Z-adjustment procedure has finished,
select **Go** (to save the results).
(The message "**Z-adjustment completed**" appears).

Mechanical adjustment Sensation 10/16/Cardiac16/40/64/Cardiac64



- Open the gantry front cover.
- Switch **S301** to "**0**".
- Open the collimator box.
- Insert the two dial gauges and adjust them to 5.00 mm.
- Release the 4 fixing screws.
- Move the collimator according to the ZAdjust program.
- Tighten the 4 fixing screws with a torque of 10.2 Nm.
- Remove the two dial gauges.
- Close the collimator box (Tighten the screws with a torque of 3 Nm).
- Switch **S301** to "**1**".
- Close the gantry front cover.



Mechanical adjustment Sensation OPEN



- Open the gantry front cover.
- Switch **S301** to "**0**".
- Secure the rotating gantry with the safety bolt.

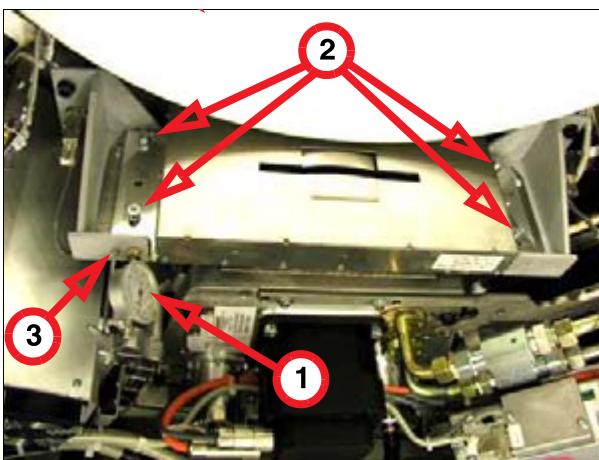


Fig. 11: Collimator adjustment



- Insert the dial gauge and adjust it to 5.00 mm (pos. 1).
- Loosen (do not remove) the 4 fixing screws (pos 2).
- Move the collimator with an Allen key according to the ZAdjust program (pos 3).
- Tighten the 4 fixing screws with a torque of 20 Nm.
- Remove the dial gauge.
- Unlock the safety bolt for gantry rotation.
- Switch **S301** to “1”.
- Close the gantry front cover.

Z position check of the tube diaphragm

General

To ensure a maximum output signal for the detector elements, the X-ray beam has to be centered precisely over the detector (Fig. 10 / p. 41). The <ZCheck> procedure provides you with a quick check of the X-ray beam alignment.

The reference value of the tube diaphragm determined with the <ZAdjust> procedure is the starting point of the quick check. Subsequently, the diaphragm is shifted symmetrically in both z-directions. Accordingly, the measured detector element output has to follow symmetrical thresholds, if the Z adjustment is still within tolerance. This procedure is automatically performed during the "Check-up" procedure at Emotions systems only. For Sensation systems the <ZCheck> is not part of the Check-up procedure.

If the result of the <ZCheck> is out of specification, please perform the <ZAdjust> procedure.

NOTE

There is no need to perform the <Z-check> procedure if you have just completed the <Z-Adjust>.

Procedure

- Select **Tune-up**.
 - Select **Expert Mode**.
 - Select **ZCheck**.
 - Select **Go**.
 - Press **START**.
- (The message "**Z-Check completed**" appears.)

Focus Alignment

General

The <FocusAli> procedure adjusts the electromagnetic focus deflection (flying focal spot) with respect to the parameters of different:

- Rotation times
- Flying focal spot modes 1x1, 1x2, 2x1 and 2x2

To determine the electromagnetic behavior of the focus deflection circuit, the following adjustments are necessary:

- **Phase:** This step determines the synchronization (delay) between the DMS and focus deflection.
- **Amplitude:** Based on the neutral position, the amplitude of the focus deflection is determined in mm. With the phase specified, the amplitude is set so that the deflection equals half the width of one detector element.
- **Offset:** With the phase and amplitude specified, the offset of the focus deflection is set so that the central beam is exactly in the center of the detector.

The procedure adjusts the settings for phase, amplitude, and offset for all combinations of these parameters

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used.[\(Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open: / p. 25\)](#)

Procedure

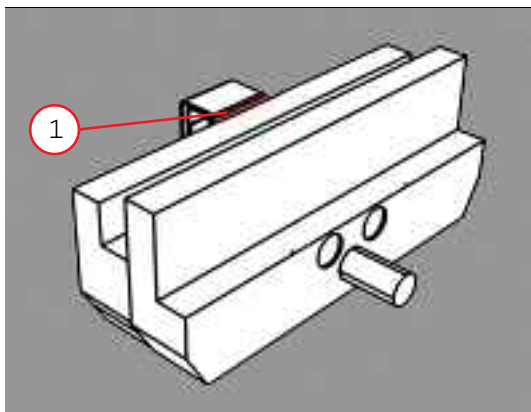


Fig. 12: Combi phantom
Pos. 1 Starting point

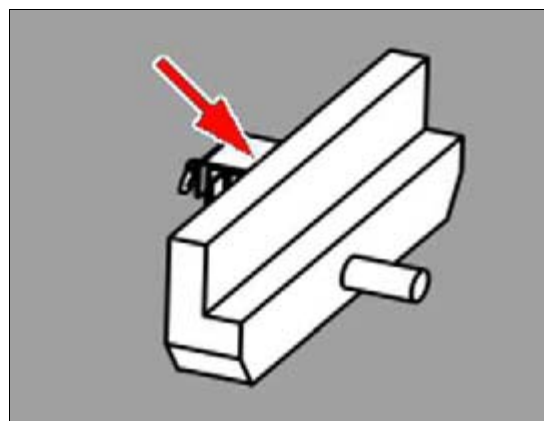


Fig. 13: Combi Phantom Sensation 64/Open

- **Combi phantom:** Position the phantom in the slice plane so that the starting point on the phantom corresponds to the light beam of the inner light marker.

or

System phantom: Position the slice thickness phantom in the measurement field so that the light beam of the inner light markers are centered above the crossover point on the slice phantom.

NOTE

With Somaris VB20 a new system phantom is available. The new system phantom contains a new high resolution phantom (wire in plexi). Somaris VB20 still supports the old system phantom (old high resolution phantom- wire in air).

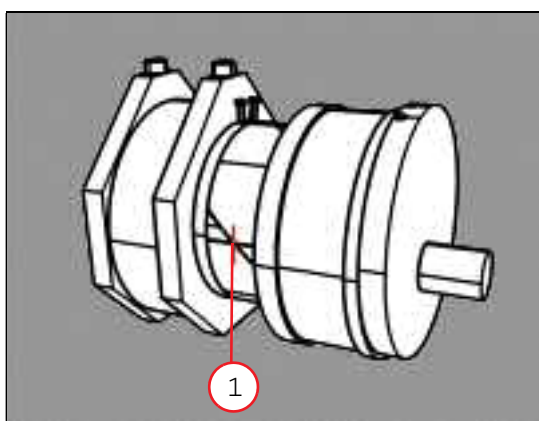


Fig. 14: System phantom
Pos. 1 Crossover point

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **FocusAli**.
- Select **Phase, Amplitude, Offset** and **Calibration**.

- Select **Go**.
- Select **combi phantom** or **system phantom**.
- Select **Go**.
- Press **START**.

- Select **Go** (to save the results).
The message "**Focus Alignment completed**" appears.

Focus Alignment Check

General

The <FocusAli> procedure adjusts the electromagnetic focus deflection (flying focal spot). The <CheckAli> procedure allows you to check the focus alignment by testing it with freely selectable parameters. CheckAli accesses the last set of raw data generated.

Scan conditions:

- Combi phantom with alignment phantom and phantom holder (see Focus Alignment).
- Mode with complete rotation was used.
- Measurement data was processed, including calibration, and stored.
- Data was stored in raw data file.

The program checks whether the correct phantom was used in the scan. If not, the alignment check procedure will be interrupted and an error message is displayed.

If the result of the <CheckAli> is out of specification, perform the <FocusAli> procedure.

Procedure

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **CheckAli**.
- Select **Fused**.
- Select the raw data file of the last measurement or any other file.
- Select **Go**.

The message "**CheckAli completed**" appears.

High Resolution UHR-plus Comb (Sensation64/Cardiac64)

General

For maximum spatial resolution, the ultra high resolution (UHR-plus) comb has to be adjusted with respect to the focal spot and the detector elements. The procedure of the <Z-CombAli> calculates the optimum Z-position of the UHR-plus comb. The procedure of the <CombAli> calculates the optimum X-position of the UHR-plus comb. The UHR-plus comb has to be mechanically shifted in the Z-direction resp. X-direction according to the results of the procedure. The UHR-plus comb is pre-adjusted at the factory.

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used.

NOTE

Disable rotation with S301 during the UHR-plus comb adjustments.

NOTE

Start with the z-adjustment first, and continue with the phi adjustment.

Procedure

Z-Adjustment

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **ZCombAli**.
- Select **Go**.
- Press **START**.
- The program informs you whether the UHR-plus comb has to be shifted.

If it is not necessary to shift the UHR-plus comb, click <GO> and wait until the message <**ZCombAli completed**> appears.

If a shift of the UHR-plus comb is required, perform the following steps:

- Loosen the two protective screw for z-direction
- Loosen the four attachment screws for UHR-Plus comb.
- Shift the UHR-plus comb with the screw "Adjustment of z-direction, as requested in the programm
- Tighten the two screws with a torque of **3 Nm**
- Tighten the four attachment screws for UHR-Plus comb.

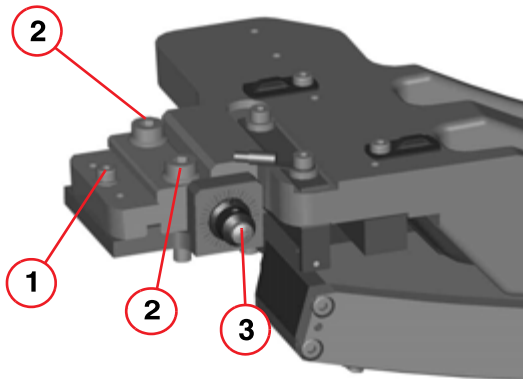


Fig. 15: UHR-plus comb, left side

- Pos. 1 Protection screw for z-direction
 Pos. 2 Attachment for UHR-plus comb
 Pos. 3 Adjustment for z-direction

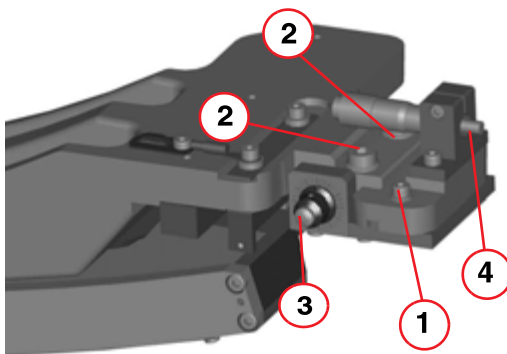


Fig. 16: UHR-plus comb, right side

- Pos. 1 Protection screw for z-direction
 Pos. 2 Attachment for UHR-plus comb
 Pos. 3 Adjustment for z-direction
 Pos. 4 Adjustment for phi- direction

Phi-Adjustment

NOTE

Note the value of the micrometer ("initial value") prior to moving the UHR-comb. If the UHR-plus comb gets completely misadjusted, the initial value can be used to restart the adjustment.

- Position the combi phantom in the slice plane so that the starting point on the phantom corresponds to the light beam of the inner light marker.

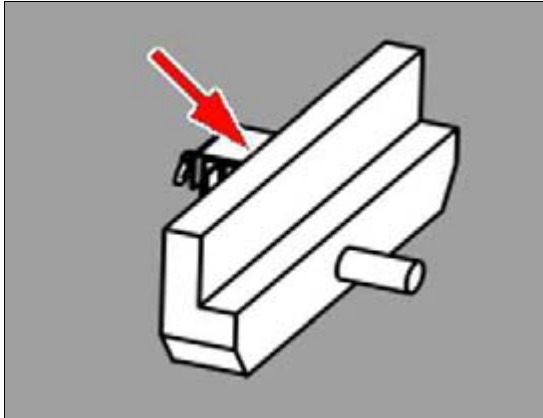


Fig. 17: Combi Phantom Sensation 64/Open

- Select **CombAli**.
- Select **Go**.
- Press **START**.
- The program informs you whether the UHR-plus comb has to be shifted.

If it is not necessary to shift the UHR-plus comb, click **<GO>** and wait until the message **<CombAli completed>** appears.

If a shift of the UHR-plus comb is required, perform the following steps:

- Make sure that the micrometer pin is attached to the UHR frame.
- Loosen the UHR-plus comb by opening 4 hexagonal screws
(2/ Fig. 15 / p. 50)(2/ Fig. 16 / p. 50)
- Shift the comb in the PHI direction according to the output of the Comb/Ali program.
Use the micrometer to move the UHR-plus comb.
One scale unit equals 0.01mm.
Turn the micrometer to the left to move the UHR-plus comb in counterclockwise direction.
Turn the micrometer the right to move the comb in clockwise direction.
- After successfully positioning, tighten the UHR-plus comb carefully using a hexagon wrench first. To assure that the UHR-plus comb does not misadjust during tightening, check the micrometer position when tightening the 4 hexagonal screws. Finally, use a torque wrench and tighten the screws with **20 Nm**.
- Continue with the CombAli program. If the UHR-plus comb is still out of range, correct the position as described before. If the UHR-plus comb is in the correct position, continue with the next step.

High Resolution Comb (All Sensations except Sensation Open)

General

For maximum spatial resolution, the ultra high resolution (UHR) comb has to be adjusted with respect to the focal spot and the detector elements. The procedure of the <CombAli> calculates the optimum phi-position of the UHR-comb. The comb has to be mechanically shifted in the phi-direction according to the results of the procedure. The UHR-comb is pre-adjusted at the factory.

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used.

Procedure

- **Combi phantom:** Position the phantom in the slice plane so that the starting point on the phantom corresponds to the light beam of the inner light marker.

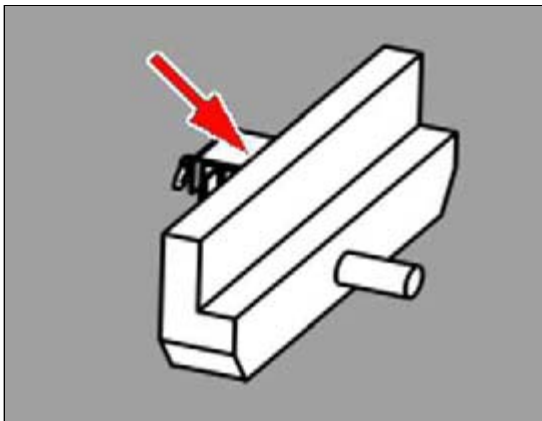


Fig. 18: Combi Phantom Sensation 64/Open

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **CombAli**.
- Select **Go**.
- Press **START**.

NOTE

Starting with VB19, the table height will be automatically adjusted throughout the procedure.

- The program informs you whether the UHR-comb has to be shifted.
If it is not necessary to shift the UHR-comb, click **<GO>** and wait until the message **<CombAli completed>** appears.

NOTE

Disable rotation with S301 during the UHR-comb adjustments.

NOTE

Note the value of the micrometer ("initial value") prior to moving the UHR-comb. If the UHR-comb gets completely mis-adjusted, the initial value can be used to restart the adjustment.

If a shift of the UHR-comb is required, perform the following steps:

- Remove both guide pinholders (**if available**) (1/Fig. 19 / p. 53) by removing the 2 x 2 screws M4 x 10mm, using a 3mm Allen key

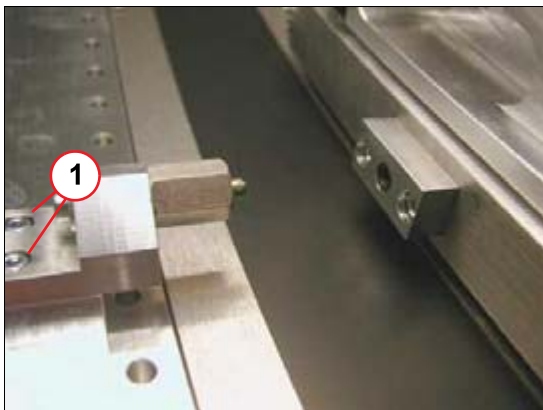


Fig. 19: UHR guide pin

Pos. 1 Screws of guide pin holder (not installed in newest systems)

- Make sure that the micrometer pin is attached to the UHR frame.
- Loosen the UHR comb by opening 4 hexagonal screws (item 1).

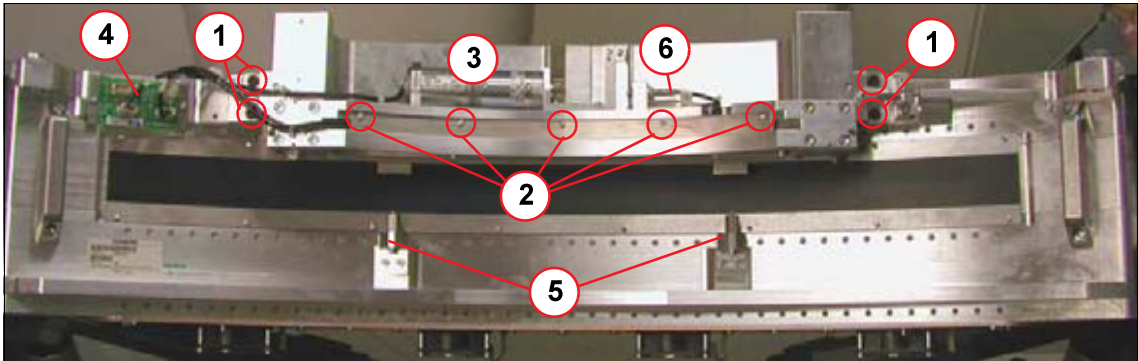


Fig. 20: UHR comb

- Pos. 1 4 screws holding the complete UHR
- Pos. 2 Screws of protective cover
- Pos. 3 UHR Motor
- Pos. 4 UHR control board
X80 is connected to FOC board D514 via W594 X5 at the C-box backplane
- Pos. 5 UHR guide pins
- Pos. 6 Limit switches (not installed in newest systems)

- Shift the comb in the PHI direction according to the output of the Comb/Ali program.
Use the micrometer to move the comb.
One scale unit equals 0.01mm.
Turn the micrometer to the left to move the comb in counterclockwise direction.
Turn the micrometer the right to move the comb in clockwise direction.

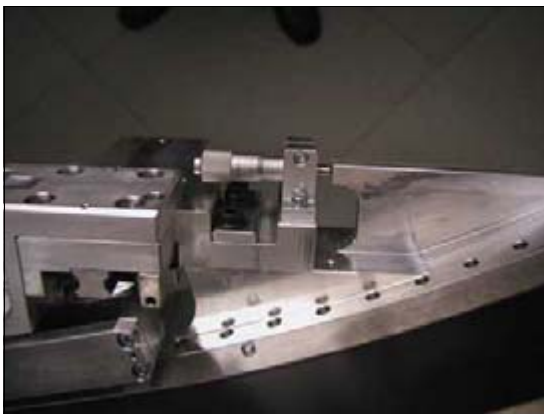


Fig. 21: UHR micrometer

- After successfully positioning, tighten the UHR comb carefully using a hexagon wrench first. To assure that the comb does not misadjust during tightening, check the micrometer position when tightening the 4 hexagonal screws. Finally, use a torque wrench and tighten the screws with **20 Nm**.
- Continue with the CombAli program. If the UHR is still out of range, correct the position as described before. If the UHR comb is in the correct position, continue with the next step.
- Adjust the UHR guide pins (**if available**) ([Fig. 19 / p. 53](#)) with the following procedure. Use service switch S1 at fiber optic cable D514 to enable the service function. To position the guide pins properly, use S3 (UHR out) and S4 (UHR in) to move the comb several times in and out the guide pins before tightening the screws of both guide pinholders with **3 Nm**

ZFocusCheck (Sensation40/64/Cardiac64, Open only since VB20)

General

This procedure checks if the beam deflection of the focal spot functions properly in the phi. and z-direction. Please be aware that this is only a check, no adjustment is possible.

Procedure

- **System phantom:** Position the slice thickness phantom in the measurement field so that the light beam of the inner light markers is centered above the crossover point on the slice phantom.

NOTE

With Somaris VB20 a new system phantom is available. The new system phantom contains a new high resolution phantom (wire in plexi). Somaris VB20 still supports the old system phantom (old high resolution phantom- wire in air).

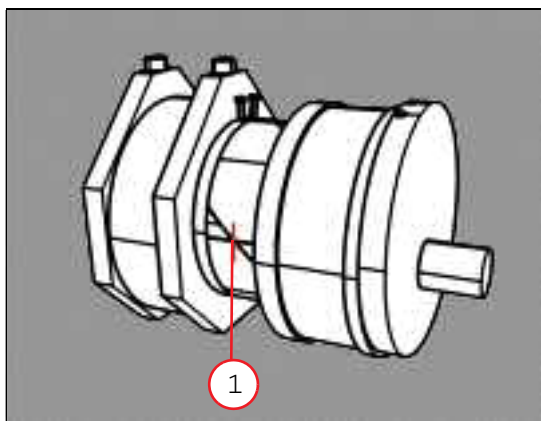


Fig. 22: System phantom

Pos. 1 Crossover point

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **ZFocusCheck**.
- Select modes: **all modes shall be selected**
- Select **Go**.
- Press **START**.

- Evaluate result:
ZFocusCheck is OK:
 - ZFocusCheck program finishes with “Function Z-FocusCheck completed” upcoming.
 - In the upcoming graphic all results for Δz are displayed in green color**ZFocusCheck is NOT OK:**
 - ZFocusCheck program finishes with “Function Z-FocusCheck completed, result(s) out of tolerance”
 - In the upcoming graphic some of the results for Δz are displayed in red color. In this case, call the CS HSC for support.
- ZFocusCheck completed

Generator adaptation

(The Plexiglas strip has to be installed.)

General

The objective of the generator adaptation procedure is to set the generator parameters (filament current, tube current, and dose) so that a specific noise value is guaranteed across the complete power (kW) range. For this reason, the noise value (sigma) is measured in a 20 cm water phantom with different power (kW) steps.

The generator adaptation has to be performed for:

- predefined kV steps

The procedure adapts the settings for all combinations of these parameters.

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used.[\(Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open: / p. 12\)](#)

Procedure

- **System phantom:** Position the slice thickness phantom in the measurement field so that the light beam of the inner light markers is centered above the crossover point on the slice phantom.

NOTE

With Somaris VB20 a new system phantom is available. The new system phantom contains a new high resolution phantom (wire in plexi). Somaris VB20 still supports the old system phantom (old high resolution phantom-wire in air).

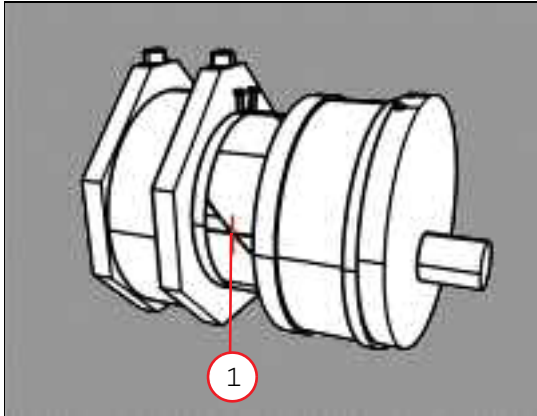


Fig. 23: System phantom

Pos. 1 Crossover point

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **GenAdapt**.
- Select voltage (kV): **select all KV steps**
- Select **Go**.
- Press **START**.
- Select **Go** (to save the results).

The message "**Generator adaptation completed**" appears.

Filament adaptation

General

The objective of the <FilAdapt> procedure is to adapt the filament current to changes in the filament due to evaporation. The filament current is adjusted to reach the X-ray dose values determined via the <GenAdapt> procedure. The generator adaptation procedure optimizes the filament current, i.e. you do not have to perform the filament adaptation procedure after you have run a generator adaptation.

This procedure is automatically added to the customer "Check-up" procedure at specific intervals.

NOTE

There is no need to perform the filament adaptation procedure if you have just completed the generator adaptation.

Required tools and phantoms

n.a.

Procedure

- Remove all absorbers from the measurement field.
 - Select **Tune-up**.
 - Select **Expert Mode**.
 - Select **FilAdapt**.
 - Select voltage (kV): **Select all KV steps**.
 - Select **Go**.
 - Press **START**.
 - Select **Go** (to save the results).
- The message "**Filament adaptation completed**" appears.

Adjustment of the Light Markers

General

Horizontal patient positioning is performed using the outer (stationary) light markers of the gantry. The marked position has to be accurately moved to the measurement field using the “Offset” button on the gantry panels. The system-specific distance between the stationary and rotating light markers is measured with a tape measure and then entered via the keyboard.

In addition, you may also use the keyboard to set and enter the table speed for movement between both light markers.

Prerequisites

Tools and auxiliary equipment

Designation	Part number
Tape measure	n.a.

Time and manpower

- 1 person
- approx. 10 minutes

Safety

WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.

CAUTION

[hz_serdoc_F04G04U14M01]

If the light marker is improperly adjusted, the applied X-radiation is not (or only partially) used.

Risk of injury!

- ⇒ Periodically check the light marker's position during “Constancy” or “Quality”.
- ⇒ If necessary, adjust the light marker according to the “Service instructions.”

Procedure

- Remove all absorbers from the measurement field.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **LightM**.
- Select **Go**.
 - ⇒ Follow the program output instructions exactly.
- To enter the "Light marker distance" in the program:
 - Click the + or - button in the parameter frame until the measured value appears
 - Directly enter the measured value into the input box the of the parameter frame
- To enter the desired table speed:
 - Click the + or - button in the parameter frame until the desired value appears
 - Use the keyboard to directly enter the desired value into the input box of the parameter frame
- Wait until the message "**LightM completed**" appears.
- Select **Go** (to save the results).

NOTE

It takes approximately 1.5 minutes to store the table and complete a new "init". During this process all gantry panel displays are dimmed.

- ⇒ The procedure is completed, if the "init" of the control unit was performed successfully. (All lights on the gantry panel displays should now be on again.)

Zero Angle Correction (Sensation40/64/Cardiac64/Open)

Refer to: [\(CT02-023.814.66 / Zero Angle Correction \(beginning with SW VB19\)\)](#)

General

Safety

⚠ WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ **Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.**

NOTE

Measurements have to be performed with the gantry doors closed.

The gap of the plexi strip has to be pointing to the one o'clock position.

However, in the case of channel correction <ChannelCorrection>, the plexi strip gap has to be pointing to the four o'clock position.

Monitor the procedure carefully for errors (streaks, rings, or other artifacts in the images).

Phantoms

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used.

Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open:

Fig. 24: Phantom holder

Overview

Table generation procedures are performed after successful completion of system-specific adjustments. Even though all adjustments are within their individual tolerance values, many software correction tables have to be generated to calculate a correct CT image acquired in different customer modes with various parameters (kV, mA, slice width, rotation time, etc.). For CT systems, the characteristics of each of the slices has be corrected as well.

Table generation includes the following:

- Warm-up
- Defective channel
- Channel correction
- Air calibration
- Water scaling
- Spacing

Save system table**Create a backup of the files after the tune-up procedure.**

Start <Backup & Restore> from the home menu of the service software

Select "Backup" in the left frame

Select a drive in the drop-down menu according to the local backup method, usually involving MOD or CD-R

Select the package "CT scanner" within the drop-down menu

Click <GO> to start the procedure

Check the screen output for the success message

Report function

Under **Local_Service --> Reports (upper navigation bar)--> Tune-Up/QA/Tests --> Tuneup** the results of the executed tuneup steps are displayed.

All tune-up steps shall be completed successfully without errors and inside the given tolerances.

If any of the tune-up routines does not finish successfully or if the result is unsatisfactory, the procedure has to be repeated.

Warm-up

(The Plexiglas strip has to be installed.)

General

The <WarmUp> procedure is used to ensure a certain minimum (X-ray tube) temperature during table generation procedures. The procedure:

- is automatically skipped if the defined minimum temperature threshold has already been reached.
- is performed automatically by all following table generation procedures if the defined minimum temperature threshold has not been reached.

Procedure

- Remove all absorbers from the measurement field.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Warm Up**.
- Select **Go**.
- Press **START**.

Wait until the message "**WarmUp completed**" appears.

Defective channel correction in Somaris VB10/VB19

General

The defective channel in Somaris VB19 (Sensation 64/Cardiac64/Open) differs slightly from version VB10. Changes are described within the section.

Basically, defective channels are handled in two different ways:

Automatic defective channel detection:

Standard procedure recommended for service purposes.

NOTE

The program “automatic defective channel detection” deletes the old automatic defective channel list and subsequently creates a new automatic defective channel list.

Channels requiring manual switch-off cannot be deleted via the automatic procedure.

The program “automatic defective channel detection” is also performed during the “checkup” procedure. However, the detected defective channels are valid only temporarily until the next “checkup” procedure is performed. No additional tune up steps are required after the “checkup” procedure.

Editing the defective channel table manually:

To switch off defective channels manually, special detector knowledge about multislice techniques is required.

Defective channel correction is performed manually only, if the defective channel cannot be detected automatically (e.g. values are too rough or problem is sporadic).

NOTE

Channels switched off manually can be reactivated only by using the same manual procedure.

Preliminary steps

- The gantry has to be closed and the system has to be in operation.
- Ensure that during table generation the system is operating under the ambient conditions specified:

If the gantry was in standby mode and rotation was inactive for a while, you should rotate the gantry for at least 10 minutes prior to generating tables.

If the gantry was switched off completely and the next step involves table generation, you should rotate the gantry for at least 20 minutes prior to generating tables.

Automatic defective channel procedure

NOTE

If a channel is marked as defective but on the next defective channel correction within tolerance, then this channel is used again. This channel uses then an “old” tuneup table, which may cause image artefacts. To avoid this, the tune-up steps “Channel Correction”, “Air Calibration”, and “Spacing” have to be performed.

Recommendation: Check for existing defects prior to run the defective channel measurement (3/4/ Fig. 25 / p. 69). Defective channels are marked in red. Perform the defective channel correction and check if previously detected channels are used again. In this case the above tuneup steps shall be performed. If all previously detected channels remain defective or even new ones are detected, no additional tune up steps have to be performed.

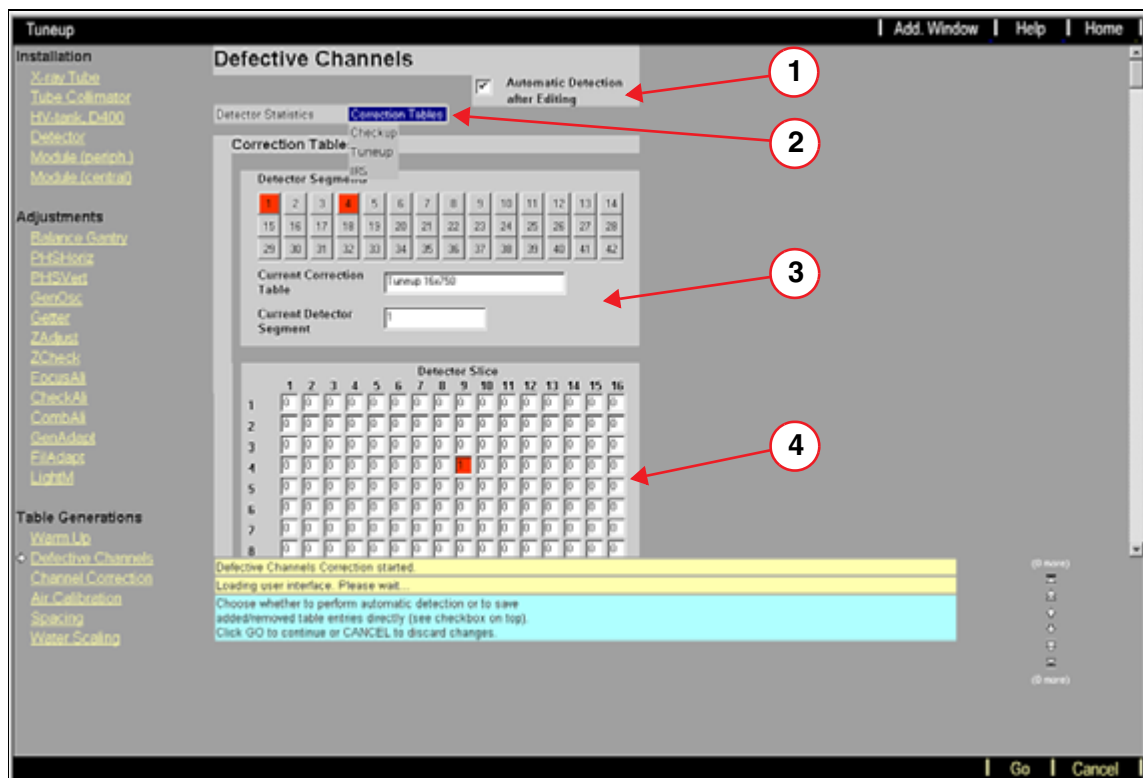


Fig. 25: Defective channels procedure

- Pos. 1 An additional automatic detection (4 measurements) may be selected. Not necessary for channel editing (patching).
- Pos. 2 Selection of Checkup, Tuneup or IRS tables.
- Pos. 3 Select a detector module for display.
- Pos. 4 The display shows the detector module selected in Pos. 3. All channels of the selected module are displayed. Defective channels are marked red (value: “1”).

- Remove all absorbers from the measurement field.
- Select **Tune-up**.

- Select **Expert Mode**.
- Select **Defective Channels**.
- **Automatic Detection after Editing** is enabled (default).
- Select **Go**.
- Press **START**.
- The results of defective channels are displayed under “Summary of Last Automatic Detection” Channels will not be marked as changed under “Affected Detector Slots”.
- Select **Go** to store the result. Prior to Somaris VB19, the checkbox “Automatic Detection after Editing” has to be removed. Otherwise, the defective channel measurement will be repeated. If you want to repeat the defective channel measurement, enable the checkbox “Automatic Detection after Editing”.
- Select **Go** to save the result.

Wait until the message "**Procedure Defective Channels completed**" appears.

Editing the defective channel table manually

NOTE

To switch off defective channels manually, special detector knowledge about multislice techniques is required.

Manual defective channel correction is performed only, if the defective channel cannot be detected automatically (e.g. values are too rough or problem is sporadic).

Channels switched off manually can be reactivated only by using the same manual procedure.

If channels are switched ON manually it might be possible that this channel causes ring artefacts because the used tuneup correction values do not fit anymore for this channel. In this case the tune-up steps “Channel Correction”, “Air Calibration”, and “Spacing” have to be performed.

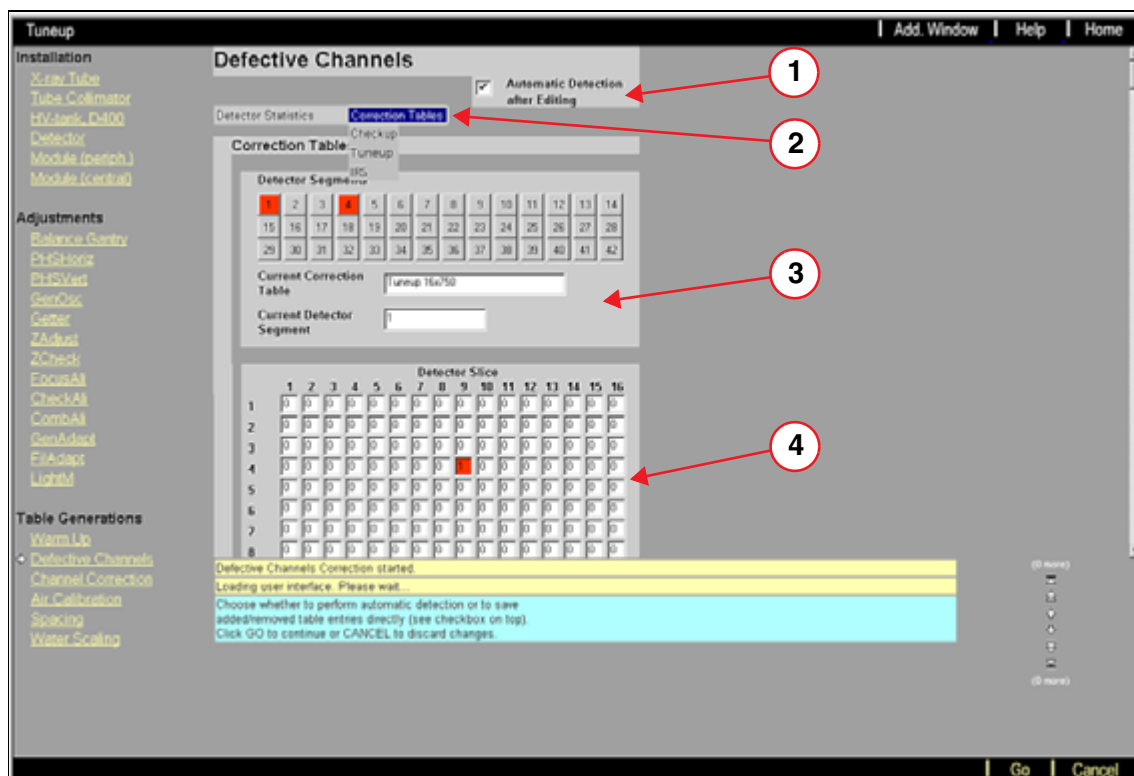


Fig. 26: Defective channels procedure

- Pos. 1 An additional automatic detection (4 measurements) may be selected. Not necessary for channel editing (patching).
- Pos. 2 Selection of Checkup, Tuneup or IRS tables.
- Pos. 3 Select a detector module for display.
- Pos. 4 The display shows the detector module selected in Pos. 3. All channels of the selected module are displayed. Defective channels are marked red (value: "1").

- Remove all absorbers from the measurement field.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Defective Channels**.
- Disable **Automatic Detection after Editing**.
- Select **Correction Tables / Tune-up**.
Select the affected **Tune-up** table (16 x 750 or 16 x 1500 for Sensation 10/16/Cardiac).
Select the affected **Tune-up** table (32 x 600 or 24 x 1200 for Sensation 64/Cardiac64/Open).
- Select the number of the **Detector Slot** with a mouse click. It may take a while before the slot number is entered under Current Detector Segment.
(Detector slots containing defective channels are highlighted in red.)

- Mark the defective channels/slices. In Somaris VB19, the detector slices are displayed from top to bottom, since the detector channels are displayed from left to right. Previous software versions displayed the above vice versa.

In Somaris VB19: Mark the channels/slices with a mouse click. Please be aware that it takes a while before the channel is marked (red color).

Before Somaris VB19: Mark the channels/slices and enter "1". To activate the setting, use the left mouse button and click anywhere in the defective channel window. The defective channels are highlighted in red.

- Select **Go**.
- Select **Go** (to save the result).

Wait until the message "**Procedure Defective Channels completed**" appears.

Detector statistics

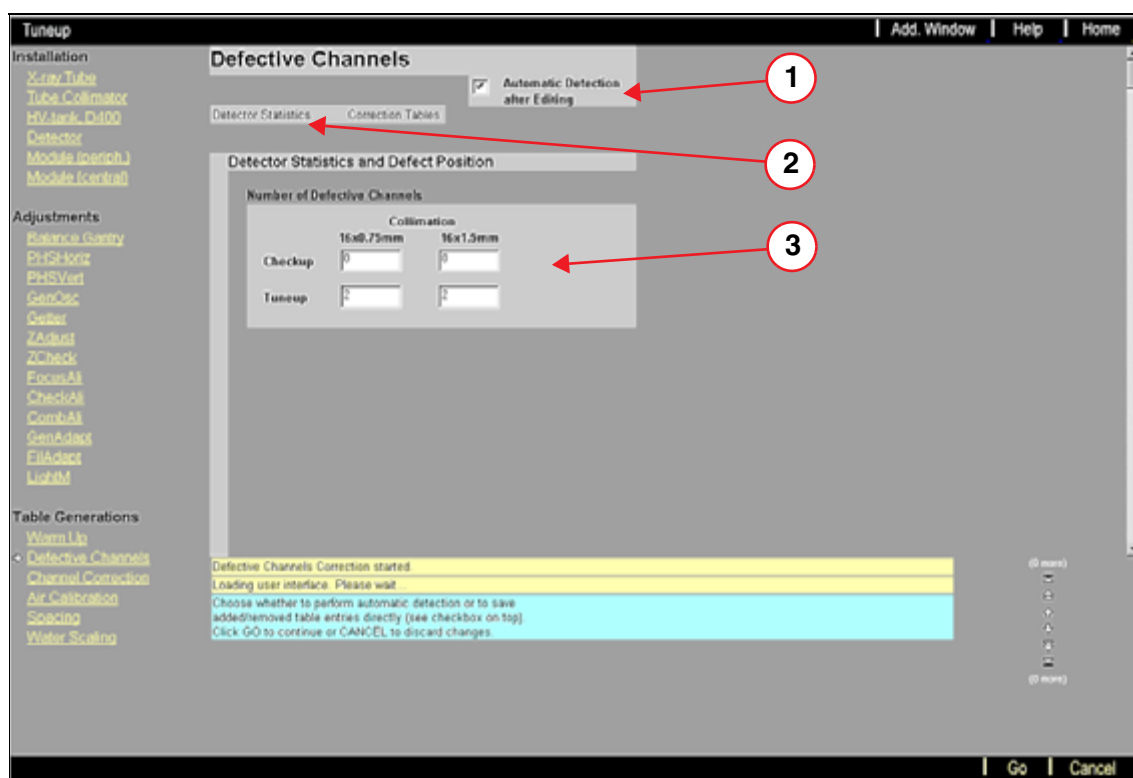


Fig. 27: Statistics of defective channels

- Pos. 1 An additional automatic detection (4 measurements) may be selected. Not necessary for channel editing (patching).
- Pos. 2 If selected, Pos. 3 will be displayed.
- Pos. 3 Displays the results of the last check-up and tune-up.

Detected defects

- The number of defective channels detected during the tune-up procedure "Defective Channels" and during the checkup can be checked under "Detector Statistics", Fig.25. In Somaris VB19, select Detector Statistics/Defect Numbers.

Defective channel history

Each channel that has ever been detected is displayed.

- Open an additional window and select **Reports**
- From the **Tune-Up/QA/Tests** menu bar, select:
Def. Chann. History

Results of defective channel measurements

A list of the defective channel measurements performed is displayed.

- Open an additional window and select **Reports**
- From the **Tune-Up/QA/Tests** menu bar, select:
Tune-up > Table Generation > Def. Channels
To view the results, double-click the report of interest.

Defective channel correction in Somaris VB20

General

The defective channel in Somaris VB20 (Sensation40/64/Cardiac64/Open) differs to previous software versions.

Basically, defective channels are handled in two different ways:

Automatic defective channel detection:

Standard procedure recommended for service purposes.

NOTE

The program “automatic defective channel detection” deletes the old automatic defective channel list and subsequently creates a new automatic defective channel list.

Channels requiring manual switch-off cannot be deleted via the automatic procedure.

The program “automatic defective channel detection” is also performed during the “checkup” procedure. However, the detected defective channels are valid only temporarily until the next “checkup” procedure is performed. No additional tune up steps are required after the “checkup” procedure.

Editing the defective channel table manually:

To switch off defective channels manually, special detector knowledge about multislice techniques is required.

Defective channel correction is performed manually only, if the defective channel cannot be detected automatically (e.g. values are too rough or problem is sporadic).

NOTE

Channels switched off manually can be reactivated only by using the same manual procedure.

Preliminary steps

- The gantry has to be closed and the system has to be in operation.
- Ensure that during table generation the system is operating under the ambient conditions specified:

If the gantry was in standby mode and rotation was inactive for a while, you should rotate the gantry for at least 10 minutes prior to generating tables.

If the gantry was switched off completely and the next step involves table generation, you should rotate the gantry for at least 20 minutes prior to generating tables.

Automatic defective channel procedure

NOTE

If a channel is marked as defective but on the next defective channel correction within tolerance, then this channel is used again. This channel uses then an “old” tuneup table, which may cause image artefacts. To avoid this, the tune-up steps “Channel Correction”, “Air Calibration”, and “Spacing” have to be performed.

Recommendation: Check for existing defects prior to run the defective channel measurement (2/ Fig. 28 / p. 75). Defective channels are marked in red. Perform the defective channel correction and check if previously detected channels are used again. In this case the above tuneup steps shall be performed. If all previously detected channels remain defective or even new ones are detected, no additional tune up steps have to be performed.

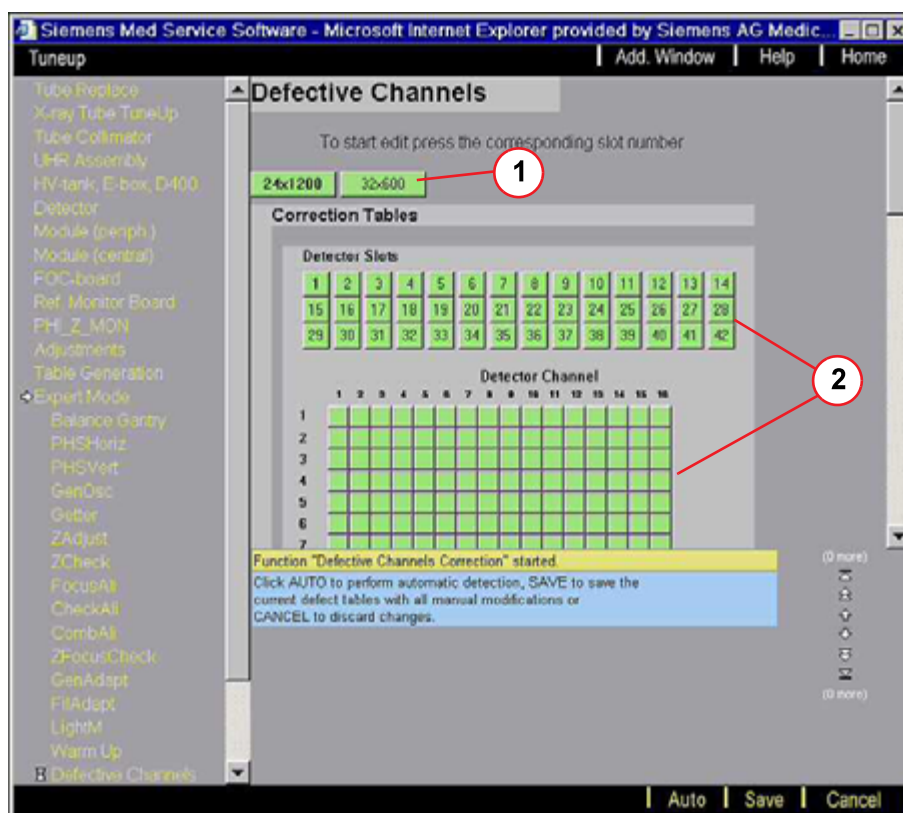


Fig. 28: Defective-Channels-VB20

- Pos. 1 Select Collimation when patching defective channels manually. At automatic detection all collimations are set as default.
- Pos. 2 When patching a channel manually, select first the affected Detector Slot and then the faulty channel / slices

- Remove all absorbers from the measurement field.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Defective Channels**.

- For automatic detection select **Auto** in the lower navigation bar
- Press **START** to perform the automatic detection of defective channels.
- Page **Defective Channels Corrections** is displayed automatically to display the results of the defective channel correction. For details refer to ([Results of Defective Channels Corrections / p. 77](#)).
- The message "**Procedure Defective Channels completed**" indicates that the procedure completed successfully.

Editing the defective channel table manually

NOTE

To switch off defective channels manually, special detector knowledge about multislice techniques is required.

Manual defective channel correction is performed only, if the defective channel cannot be detected automatically (e.g. values are too rough or problem is sporadic).

Channels switched off manually can be reactivated only by using the same manual procedure.

If channels are switched ON manually it might be possible that this channel causes ring artefacts because the used tuneup correction values do not fit anymore for this channel. In this case the tune-up steps "Channel Correction", "Air Calibration", and "Spacing" have to be performed.

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Defective Channels**.
- Select the **Correction Table** (e.g. 24x1200) which shall be patched manually ([1/Fig. 28 / p. 75](#)).
- Mark the affected **Detector Slot** by selecting the number with a mouse click ([2/Fig. 28 / p. 75](#)). The color changes to red indicating that this Detector slots contains defective channels.
- Mark the defective channels and the affected slices by selecting the channel number in all affected slices with a mouse click ([2/Fig. 28 / p. 75](#)). Detector slices are displayed from top to bottom, detector channels are displayed from left to right. Defective detector channels are highlighted in red.
- Select **Save**.
- Page **Defective Channels Corrections** is displayed automatically to display the results of the defective channel correction. For details refer to ([Results of Defective Channels Corrections / p. 77](#)).

Results of Defective Channels Corrections



Fig. 29: Result -Defective-Channels-VB20

- Affected Modules: Displays the module to be exchanged
- Reference Detector: Displays the status of the reference detector
- Table Editing History: Displays the result of the last defective channel measurement (automatic or manual)
- Detector Statistics: Displays all defective channels (automatically or manually detected) with the affected collimation.
- Summary of Last Automatic Detection: Displays the result of the last performed automatic defective channel correction.

Defective channel history

Each channel that has ever been detected is displayed under

- Local_Service --> Reports --> Tune-Up/QA/Tests --> Def. Chann. History

Results of defective channel measurements

A list of the defective channel measurements performed is displayed under

- **Local_Service --> Reports --> Tune-Up/QA/Tests --> Tune-up --> Table Generation --> Def. Channels** (to view the results, double-click the report of interest.

Channel Correction

(The Plexiglas strip has to be installed.)

General

The <ChannelCorrection> procedure compensates for the difference in attenuated signals from detector element to detector element. The correction has to be performed for different scan parameters (e.g. kV, slice width, etc.) as well as for the different densities of the absorbers and the path covered within an absorber (Sensation 10/16/Cardiac only). The channel correction performed for Sensation 10/16/Cardiac requires head and body absorbers as Sensation 64/Cardiac64 and Open only require the body absorber to generate the correction table. The head correction tables are generated from the body tables.

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used. [\(Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open: / p. 25\)](#)

Procedure

NOTE

Only in the case of channel correction <ChannelCorrection>, the plexi-strip gap has to point to the four o'clock position.

When performing Channel Correction <ChannelCorrection> only, ensure that the gantry was rotating for at least 10 minutes prior to the procedure.

- Move the plexi strip gap to the 4 o'clock position

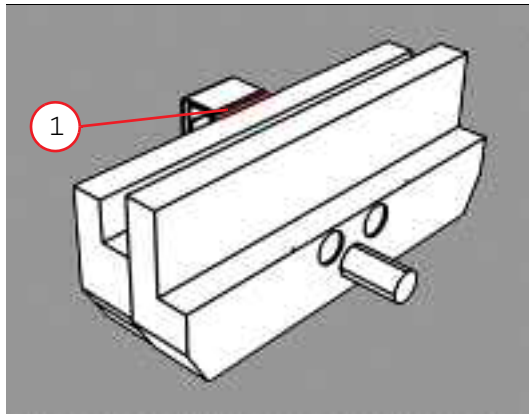


Fig. 30: *Combi phantom*
Pos. 1 Starting point

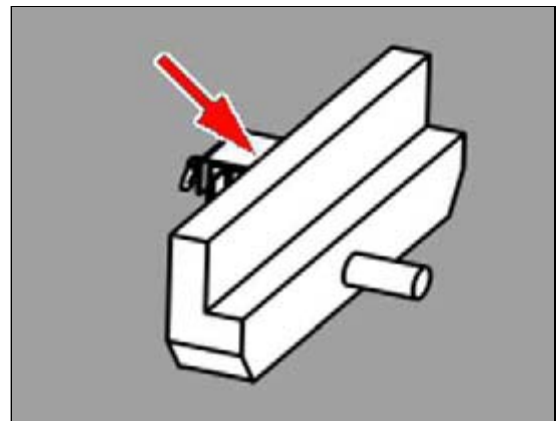


Fig. 31: *Combi Phantom Sensation 64/Open*

- **Combi phantom:** Position the phantom in the slice plane so that the starting point on the phantom corresponds to the light beam of the inner light marker.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Channel Correction**.
- Switch on **check** (default, a check measurement is performed).
- Click **select all** (all Body and Head Modes are selected, for Sensation64/Cardiac64/Open only body modes are selected).
- Select **Go**.
- Press **START**.
Wait until the message "**Channel Correction completed**" appears.
- Move the plexi strip gap to the 1 o'clock position

Air calibration

(The Plexiglas strip has to be installed.)

General

The channels of the measurement system have different levels of sensitivity that are compensated for by <Air Calibration>. Air calibration consists of two major steps, base calibration and difference calibration.

The objective of base calibration is to determine the changes in sensitivity caused by present environmental conditions. For this reason, base calibration has to be performed on a daily basis. This function is automatically selected when the system is switched on. In addition, the customer can run base calibration under "Setup/Calibration", as required.

The objective of the difference calibration procedure is to determine the structural differences in sensitivity for the various scan modes (kV, mA, slice width, rotation time, etc.). For this reason, difference calibration tables are generated during tune-up only.

Air calibration checks whether an absorber is in the X-ray beam. If the message indicates "There may be an object in the scan field..." although no absorber is in the X-ray beam, the reason could be contrast on the plexi strip, a defective channel, or the adjustment of the collimator box (Z-Adjust).

Procedure

- Remove all absorbers from the measurement field.
- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Air Calibration**.
- Click **all modes**.
- Select **Go**.
- Press **START**.

Wait until the message "**Air Calibration completed**" appears.

Spacing correction

(The Plexiglas strip has to be installed.)

General

This procedure is used to correct the mechanical positions of the single detector elements. These so-called "spacing errors" can result in visible artifacts in tomograms.

The <Spacing> correction mathematically equalizes the different distances of the single detector elements. A spacing correction table is created to compensate for spacing errors during image reconstruction.

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used. ([Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open: / p. 25](#))

Procedure

- **Combi phantom:** Position the phantom in the slice plane so that the starting point on the phantom corresponds to the light beam of the inner light marker.

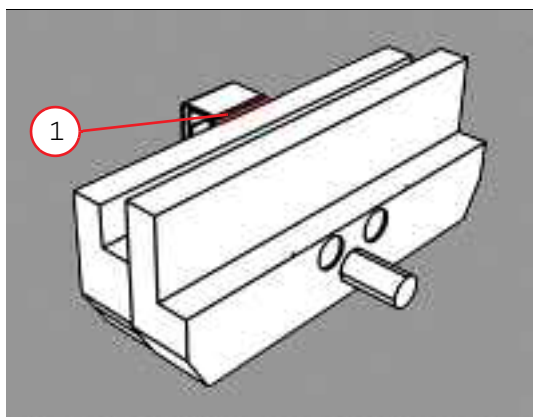


Fig. 32: Combi phantom
Pos. 1 Starting point

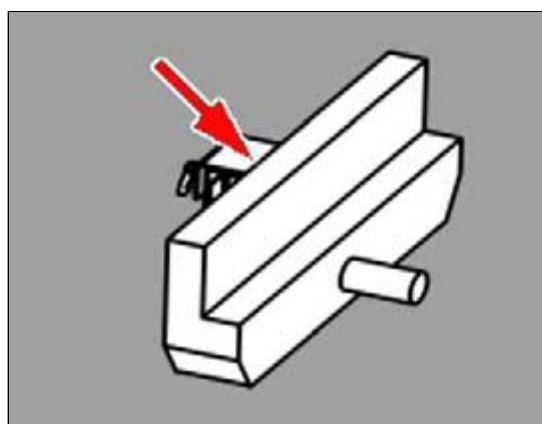


Fig. 33: Combi Phantom Sensation 64/Open

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Spacing**.
- Select **Go**.

- Press **START**.

Wait until the message "**Spacing completed**" appears.

Water scaling

(The Plexiglas strip has to be installed.)

General

The density of water is defined as 0 HU on the Hounsfield scale. To ensure that the system meets this requirement, each measurement value is multiplied by a scaling factor during pre-processing. These scaling factors are determined via the <WaterScaling> procedure function. The scaling factors have to be generated for each scan parameter (kV, slice width, etc.).

NOTE

The density of water is a function of its temperature. A difference of +10 Kelvin (18 degrees Fahrenheit) results in a difference of -3 HU on the CT scale. For reproducible scaling factors, the water temperature inside the phantom has to be the same for all <WaterScaling> procedures (QUALITY/CONSTANCY). For this reason, it is recommended that the water phantom be stored in the scan room to ensure a constant water temperature level.

Approx. 5 hours are required for the water in the phantom to compensate for a 10 Kelvin difference in temperature (18 degrees Fahrenheit).

NOTE

Only the original phantoms (delivered with the system) can be used.

The vertical and horizontal positioning of the phantom is performed automatically at a predefined table height. For this reason, only the original phantom holder (delivered with the system) can be used. ([Phantom holder for Sensation 10/16/Cardiac16/40/64/Cardiac64/Open: / p. 25](#))

Procedure

- **System phantom:** Position the slice thickness phantom in the measurement field so that the light beam of the inner light markers is centered above the crossover point on the slice phantom.

NOTE

With Somaris VB20 a new system phantom is available. The new system phantom contains a new high resolution phantom (wire in plexi). Somaris VB20 still supports the old system phantom (old high resolution phantom-wire in air).

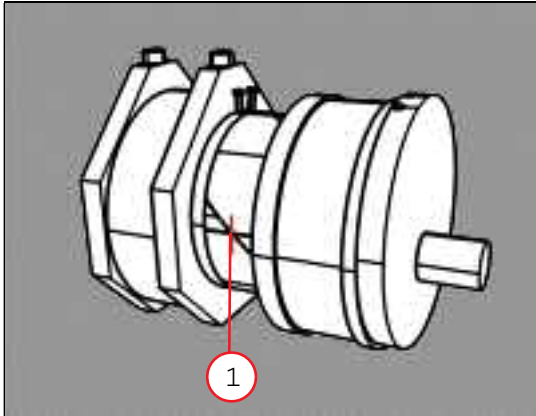


Fig. 34: System phantom

Pos. 1 Crossover point

- Select **Tune-up**.
- Select **Expert Mode**.
- Select **Water Scaling**.
- Click **select all** (modes).
- Select **Go**.
- Press **START**.

Wait until the message "**Water scaling completed**" appears.

Final Work

Screen shot (VB10)

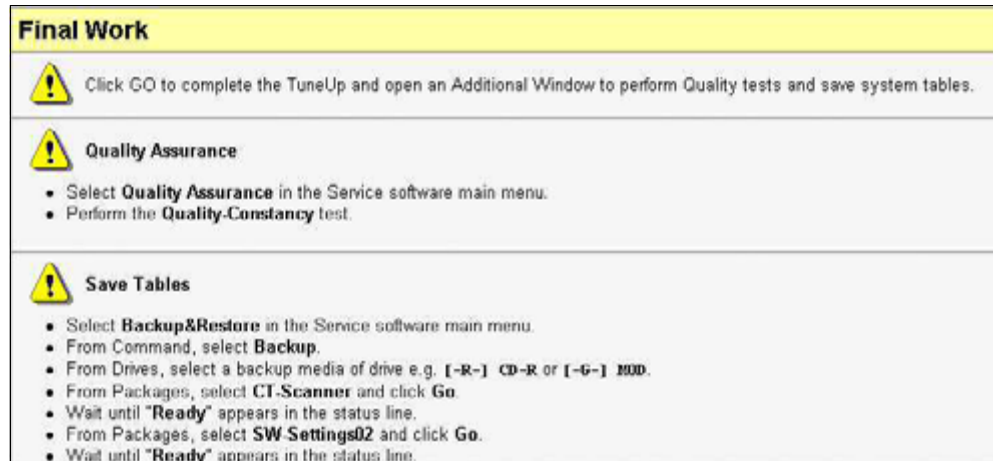


Fig. 35: Final work

Quality Assurance

- Select Quality Assurance in the Service software main menu.
- Perform the Quality Constancy test.

Backup

- Select Backup & Restore in the Service software main menu.
- From Command, select **Backup**.
- From Drives, select a backup media or drive, e.g. **[-R-] CD-R** or **[-G-] MOD**.
- From Packages, select **CT-Scanner** and click **Go**.
- Wait until "Ready" appears in the status line.
- From Packages, select **SW-Settings02** and click **Go**.
- Wait until "Ready" appears in the status line.

Chapter	Change	Reason
3	“Balancing the gantry” Notice added	CTTU-05-0010

H

hz_serdoc_F04G04U14M01	61
hz_serdoc_F13G01U03M01	6
hz_serdoc_F13G01U12M03	12, 25, 27, 32, 37, 61, 64
hz_serdoc_F13G01U15M01	7
hz_serdoc_F13G10U01M01	7

